ZCAS University

CCS1601 INTERNET OF THINGS FUNDAMENTALS

FINAL EXAM

31ST MAY 2024

Time: 12:30 – 15:30 Hours

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

Dr. Evelyn Chewe, a renowned plant pathologist at ZCAS University, found her frustration mounting. Her research on a new, virulent strain of Fusarium wilt threatening her tomato crops had stalled. Traditional methods of monitoring greenhouse conditions – manual temperature and humidity checks, and laborious soil moisture analysis – were time-consuming and prone to human error. Valuable data points were missed, hindering her ability to pinpoint the optimal environment for resisting the fungal infection.

One evening, while lamenting her predicament to a colleague, Dr. Chewe was introduced to a cutting-edge innovation – the "Verdant Guardian" smart greenhouse system. This system, developed by a team of engineers and agricultural scientists, promised a comprehensive, data-driven approach to greenhouse management.

Intrigued, Dr. Chewe secured funding and wants to install the Verdant Guardian for her greenhouse. The system boasts of a network of meticulously positioned sensors – temperature, humidity, soil moisture, and even ambient light levels. These sensors fed data into a central hub; a sleek AI-powered unit dubbed "Verdant Mind."

Verdant Mind, Dr. Chewe soon discovered, was the brain of the operation. It not only monitored and visualized real-time data but also offered automated control over ventilation, irrigation, and even supplemental lighting. Dr. Chewe can program the system to mimic specific microclimates or run experiments with precise environmental adjustments.

The information was nothing but astonishing. Dr. Chewe, for the first time, had a complete picture of her greenhouse environment. She could identify micro-fluctuations in temperature and humidity that might have gone unnoticed before. By meticulously analysing Verdant Mind's data and adjusting environmental factors accordingly, Dr. Chewe identified a specific temperature and humidity range that significantly inhibited the growth of the Fusarium wilt fungus. This could be the breakthrough that can pave the way for developing targeted treatments and preventative measures to protect tomato crops across Zambia, Dr Chewe pictured in her mind.

You are tasked with helping Dr. Chewe set up this IoT system for her smart greenhouse that can automatically maintain optimal climate conditions for growing tomatoes that she has been introduced to.

(a) Describe how will you connect the sensors and actuators to the microcontroller. Explain the wiring and communication protocols involved.

(5 marks)

(b) Explain how the microcontroller will connect to the internet. What factors will influence your choice of connectivity method?

(5 marks)

(c) Name the specific types of sensors will you use for temperature, humidity, and soil moisture. How will the collected sensor data be interpreted by the microcontroller?

(5 marks)

(d) Discuss how the microcontroller controls the actuators (exhaust fan, dehumidifier, solenoid valve) to maintain optimal temperature, humidity, and soil moisture levels for tomatoes based on the sensor data. (5 marks)

(e) Explain the role of the microcontroller in this system. What programming language will you use to write the code that controls the entire system based on sensor data. (5 marks)

(f) Discuss the benefits of using a cloud platform for data storage and remote monitoring of the greenhouse conditions. (5 marks)

(g) Write a python program that can group multiple variables together like in a container. Show the output and add any comments (5 marks)

(h) Define the term Indexing in Python programming language (2 marks)

(i) Explain the terms "***Analog Circuits***" and "***Digital Circuits***." (3 marks)

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

a) Shown below is a code that prints a list of fruits. Write a code using a FOR LOOP to INDEX the list of fruits. Show your output

```
fruits = ["apple", "orange", "tomato", "banana"]  # a list of strings
print(type(fruits))
print(fruits)
```

```
<class 'list'>
['apple', 'orange', 'tomato', 'banana']
```

(6 marks)

b) Write Python code that executes TEN numerical values and squares them (8 marks)

c) Write a program of your choice using a WHILE LOOP (6 marks)

(20 marks)

Question 3

Electronics is a field of study focused on the control of electricity and the physical components and circuits that help direct electrical energy.

(a) Draw a basic circuit diagram (schematic) showing the current, resistor, and LED.

(5 marks)

(b) Define cloud computing and describe the user benefits of IaaS, PaaS, mPaaS, SaaS

(10 marks)

(c) Explain what the Rasberry Pi and Arduino board is used for in IoT (5 marks)

(20marks) Question 4

Functions allow us to package functionality in a nice and readable way, reuse it without writing it again, and make code modular and readable, Rule of thumb - if you are planning on using very similar code more than once, it may be worthwhile writing it as a reusable function.

(a) Write a Python program using the following functions showing indentation.

i. def (5 marks)

ii. if else (5 marks)

iii. break (5 marks)

(b) Write a Python program the defines a dictionary using the type (days of the week)

(5 marks)

(20 marks)

Question 5

Cloud customers have access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort. Discuss the following.

a) The Fog Computing Model (5marks)
b) Big Data and Security concerns in IoT (5 marks
c) Data Transmission (5 marks)
d) Cloud Computing (5 marks)

(20 marks)

END OF EXAM